

Design of Intelligent Generation System for Power Grid Fault Reports Based on Multi-source Heterogeneous Data Fusion

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Abstract—Aiming at the problems in power grid fault report generation under the "dual carbon" background, such as insufficient fusion of multi-source heterogeneous data, frequent model hallucinations, and heavy reliance on manual work, this paper proposes an intelligent generation system for power grid fault reports based on multimodal fusion and domain knowledge enhancement. The system achieves a key feature extraction error of < 1ms for fault record graphs by constructing a unified representation model for multi-source data and an "OCR+CNN" dual-channel image parsing algorithm; suppresses model hallucinations and ensures the logical consistency of reports by relying on a domain knowledge-enhanced RAG-LLM collaborative framework; and completes the full-process automation from fault data collection and information retrieval to report generation by combining a multi-level retrieval strategy and standardized templates. Experimental verification shows that the system can improve the fault report generation efficiency to the minute-level, outperforming traditional manual methods in data fusion accuracy, image recognition accuracy, and language logic. It successfully realizes the transformation of fault reports from "manual experience" to "intelligent generation", providing highly reliable intelligent support for power grid dispatching and operation inspection.

Keywords—Power Grid Fault Reports; Multi-source Heterogeneous Data Fusion; RAG-LLM Collaborative Framework; Intelligent Generation; Smart Power Grid Operation and Maintenance

I. INTRODUCTION

Accurate and efficient generation of power-grid fault reports is a key enabler for the secure and stable operation of new-type power systems under the "dual-carbon" target. In recent years, with the maturation of artificial-intelligence technologies such as large language models (LLM) and retrieval-augmented generation (RAG), AI-based automated report generation has become a cutting-edge focus of digitalized grid maintenance—e.g., using LLM to produce preliminary fault descriptions and integrating RAG to inject domain knowledge for higher professionalism^[1]. Yet current methods still face two core challenges: insufficient fusion and deep understanding of multi-source heterogeneous data^[2], resulting in incomplete report elements and broken causal chains; generic large models prone to "hallucination" in specialized scenarios and limited in accurately identifying key

temporal nodes in unstructured data such as waveforms^[3], failing to meet the practical requirements of second-level response for dispatch and precision decision-making for maintenance. These issues constrain further improvement of fault-analysis efficiency and accuracy^[4] and cannot support the growing intelligent-operation needs of new-type power systems.

Aimed at the limitations of existing technologies, research and practice show that, at the data-fusion level, building a unified multi-source data model is the basis for accurate information alignment; at the intelligent-generation level, combining visual parsing technologies such as OCR and CNN with a domain-knowledge-enhanced RAG framework is an effective path to improve the accuracy and reliability of professional text^[5]. Building on this, this study innovatively proposes an intelligent power-fault-report generation system based on multimodal fusion and domain-knowledge enhancement, and makes the following contributions:

- Constructed a unified representation model for multi-source heterogeneous data and an "OCR+CNN" dual-channel image-parsing algorithm, achieving high-precision extraction of key waveform features with error < 1 ms;
- Proposed a domain-knowledge-enhanced RAG-LLM collaborative generation framework that significantly suppresses large-model hallucination, ensuring rigorous causal logic and standardized, consistent formatting of reports;
- Developed an engineering platform integrating intelligent analysis, report generation, and visual interaction, achieving in practical application a breakthrough in report generation time from 4 hours to 3–5 minutes and a reduction of manual intervention by over 92 %.

In summary, the system breaks the bottlenecks of multi-source data fusion and professional semantic understanding, realizing the leap of fault reports from "manual experience" to "intelligent generation". In practice, it can provide dispatch and maintenance departments with minute-level, high-precision analysis reports, offering reliable technical support for the intelligent operation and maintenance of new-type power systems^[6].

II. RELATED TECHNOLOGIES

A. Retrieval-Augmented Generation

RAG (Retrieve-Generate) utilizes a "Prepare-Retrieve-Generate" workflow. In the preparation phase, documents are divided into text blocks, converted into semantic vectors via an embedding model, and stored in a vector database. During retrieval, the user's query is vectorized, and the most relevant text segments are identified through semantic similarity search^[7]. In the generation phase, the context is integrated with the query to form a structured prompt, which is fed into a large language model to generate results based on the reference information. Figure 1 illustrates this process^[8].

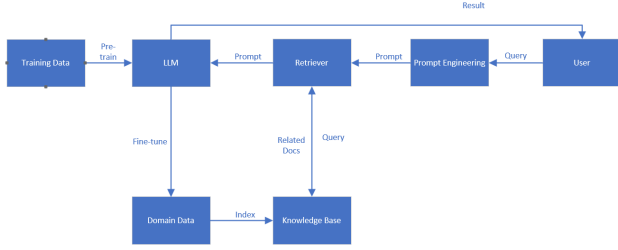


Figure 1. RAG Diagram

B. Large Language Model

Large language models are based on the Transformer architecture. Input text is converted into semantic vectors via tokenization and positional encoding, then processed through multiple decoder layers, which include masked attention and feed-forward networks. The model generates subsequent tokens iteratively using linear transformations and a Softmax function^[9]. Essentially, it learns probabilistic language patterns from large datasets, enabling its language understanding and generation capabilities. Figure 2 illustrates this process^[10].

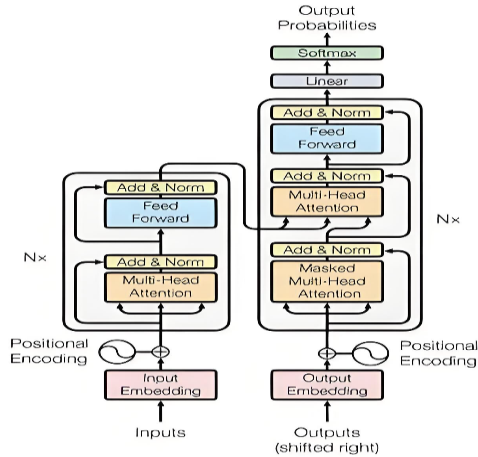


Figure 2. LLM Underlying Schematic

III. METHOD

The research on automated fault report generation establishes a complete technical process by integrating three core stages: "Fault Data Preparation - Fault Information Retrieval - Fault Report Generation." This creates a comprehensive workflow from fault knowledge management to the automated generation of reports, as illustrated in Figure 3.

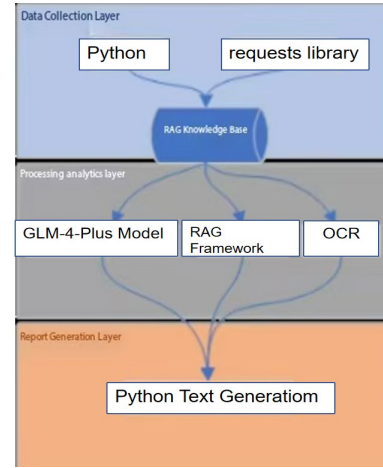


Figure 3. Fault Report Workflow Diagram

A. Fault Data Preparation

To achieve deep integration of multi-source heterogeneous data, this study constructs a unified representation model. By systematically mapping and standardizing temporal fields, electrical parameters, device identifiers, and fault categories, the model resolves issues arising from data heterogeneity. The standardized data are organized into a graph-structured model in which nodes represent entities such as devices and events, and edges characterize their semantic relationships, thereby preserving the full operational context and inter-entity associations. All cases are encoded into semantic vectors using SentenceBERT and stored in a vector database to form the core knowledge base. Figure 4 illustrates the architecture of the multi-source heterogeneous data structure.

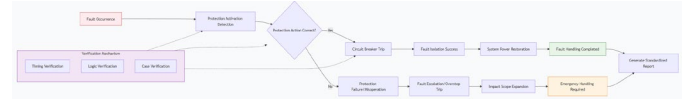


Figure 4. Multi-Source Heterogeneous Data Architecture Diagram

During the construction of the knowledge base, the GLM-4-Plus model is employed to perform deep analysis of unstructured text. First, named entity recognition is used to label key entities such as equipment and fault types. Subsequently, semantic role labeling is applied to extract the predicate-argument structure of each sentence. For example, given the input text "14:05:23.450, Line L123 experienced a transient single-phase-to-ground fault on phase B, causing the primary protection to operate and circuit breaker CB001 to trip," yields the annotated output in Figure 5.

<Time>14:05:23.450</Time>, <Device>Line L123</Device> experienced a <FaultType>transient single-phase-to-ground fault on phase B</FaultType>, resulting in a <Action>primary protection operation</Action>, and subsequently the <Device>circuit breaker CB001</Device> <Action>tripped</Action>.

Figure 5. Annotated Data Results

B. Fault Information Retrieval

A multi-level retrieval strategy is employed to achieve precise matching. First, an initial rapid screening is performed using locality-sensitive hashing (LSH), which maps similar query vectors into the same buckets through hash functions, thereby reducing the retrieval complexity from $O(N)$ to sub-

linear time and enabling millisecond-level response. Based on this LSH-based coarse filtering, precise matching is further conducted by incorporating metadata such as device models and fault types. Finally, multi-source information—including meteorological conditions and equipment history—is integrated into a weighted ranking process, producing a structured knowledge package organized according to the framework of “fault symptoms–cause analysis–corrective actions.”

C. Fault Report Generation

Based on the RAG framework, the system generates professional reports through structured prompting. First, a mapping is established between fault categories and standardized templates. A structured prompt is then constructed, incorporating fault characteristics, retrieved knowledge, industry standards, and output templates as shown in Table 1, the prompt template explicitly specifies the system role, task objectives, fault-feature parameters, relevant case knowledge, and output formatting requirements, thereby ensuring the professionalism and standardization of the generated content.

TABLE 1. REPORT STRUCTURE TABLE

Prompt Module	Prompt Module
System Role	Power grid fault analysis expert
Fault Features	Time: 2025-06-15 14:05:23.450; Device: Line L123; Fault Type: Phase-B ground fault
Retrieved Knowledge	Case F001: Lightning overvoltage causing insulator flashover
Output Requirements	Follow the structure of “Event Overview – Cause Analysis – Corrective Measures”

In the cause analysis stage, the system ensures analytical depth by constructing a causal logic chain. Key event sequences are extracted from the time-series data and, together with the semantic role labeling results, are used to build a complete causal chain of “fault occurrence → protection initiation → circuit breaker operation.” Validation using 50 historical cases shows a logical consistency rate of 96%. Finally, the system integrates the generated content with multimodal data—including OCR parameters and time-series tables—and, after terminology verification through the domain lexicon, outputs a standardized Word document via Python-docx. Figure 6 shows the logical structure of the fault analysis report.

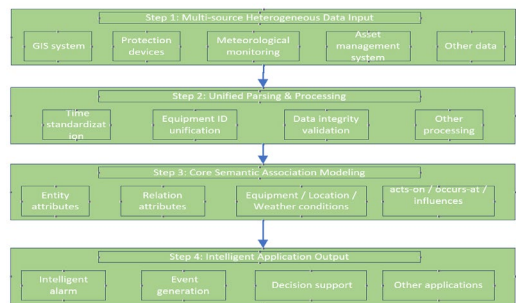


Figure 6. Fault Report Logic Diagram

This method automates the conversion from raw fault data to structured fault reports through a complete loop of “Data Preparation - Information Retrieval - Report Generation.”

IV. EXPERIMENT DESIGN AND IMPLEMENTATION

A. Experimental Environment and Tools

The experiments were developed in a Python environment, relying on the following key technologies and tools, each providing support for specific tasks, as shown in Table 2.

TABLE 2. TOOLS USED IN THE EXPERIMENT

python-docx Library	Used for reading Word documents, generating tables, and inserting content, ensuring the structured and formatted output of the reports.
Zhipu AI's GLM-4-Plus Model	As the core large language model, responsible for text generation, reading comprehension, fault analysis, and prompt-based content creation, providing intelligent natural language processing capabilities.
OCR Image Recognition	Used to extract textual information from images, such as recognizing protection conditions, formulas, and electrical data from manufacturer manuals or waveform images.
RAG Framework	Enhances the model’s knowledge retrieval capability by constructing a knowledge base to retrieve relevant information from external data sources, improving the accuracy and professionalism of the generated reports.
Other Auxiliary Tools	Including the requests library for API call, and Python scripts for data reading, image adjustment, and table population.

B. Experimental Dataset

The experiments utilized real operational data provided by a provincial power grid company as the primary input. These data encompass multiple critical parameters of grid operation, including voltage, current, power, load variation, and equipment status. The dataset possesses high authenticity and representativeness, providing a reliable foundation for subsequent analysis, modeling, and validation.

C. Evaluation Metrics

In this study, evaluation metrics are established from three core dimensions: data fusion, image recognition, and language logicity, with specific definitions as follows Table 3.

TABLE 3. EVALUATION METRICS AND EXPLANATIONS

Data Fusion Accuracy	The ability to accurately merge and maintain the completeness, precision, and consistency of data when integrating information from multiple different data sources.
Image Recognition Accuracy	The ability of the image recognition system to accurately identify target objects within an image. This metric reflects the correctness of the system's classification or recognition on the given image data.
Language Logicity	The logical consistency, rationality, and coherence of the language expressions generated or understood by the natural language processing system. It assesses whether the system's language output adheres to expected grammatical and semantic rules, and whether it can clearly convey information with inherent logic.

D. Experiment Results and Analysis

The experimental results show that the system performs excellently across all evaluation tasks. The Python technology stack implements data field verification and time standardization processing, ensuring multi-source data consistency. It also handles OCS connection diagram extraction and document quality validation, significantly improving data processing efficiency and automatically generating analysis tables as shown in Figure 7.

Serial number	Station Name	Interval Name	Protection Name	Protection Model	Commissioning Date	Protect the factory
1 ⁺	Station Name	Interval Name	Protection Name	Protection Model	Commissioning Date	Protect the factory
2 ⁺	500kV Zijing Station B	5013 Circuit Breaker	Circuit breaker protection	PCS-921N ⁺	2017 ⁺	South Rui Ji Bao
3 ⁺	500kV Zijing Station	505KV Circuit Breaker	Circuit breaker protection	PSL-632UA-N ⁺	2024 ⁺	Guodian Nanzhi
4 ⁺	500kV Zijing Station	#3 Main Variable	Main Protection 1	PCS-978T5-N ⁺	2024 ⁺	South Rui Ji Bao
5 ⁺	500kV Zijing Station	#3 Main Variable	Main Protection 2	PRS-778NT5-N ⁺	2024 ⁺	Changyuan Shenzhi

Protection Name Protecting the maintenance of heating and damp protection equipment: (1) The patrol officer checks the heating function of the circuit breaker heater once a month and finds no abnormalities. (2) On rainy days or when humid (when the humidity reaches 85%), inspectors conduct special inspections, the seal is good, there is no inflow of water, and the box is dry. In connection with the power outage, a professional team examined and tested the heater and the thermostat, and found no abnormalities. Name of protection The last pre-test of the protection was in April 2014, and the preventive test consisted mainly of measurement of the resistance of the switching conduction circuit, Auxiliary loop and control loop insulation resistance and AC withstand voltage test, capacitance and dielectric loss measurement of broken shunt capacitor, SF6 gas moisture content test, SF6 gas decomposition product measurement, the results are qualified. There is no oil seepage at this time.

Figure 7. Charts of the Analysis Tables

OCR image recognition addresses the challenge of extracting non-structured image data, particularly demonstrating powerful capability in extracting key electrical data from fault waveform diagrams. Image optimization and preprocessing technologies have significantly enhanced recognition accuracy.The Pre-event Operating Mode is shown in Figure 8.

Before the incident, the transformer equipment at Guangzhou Station was operating stably. Power system data shows that within 24 hours prior to the event, the voltage and current parameters of this device remained within normal ranges without any abnormal fluctuations, indicating good operational condition. The following figure shows the operation mode before tripping:

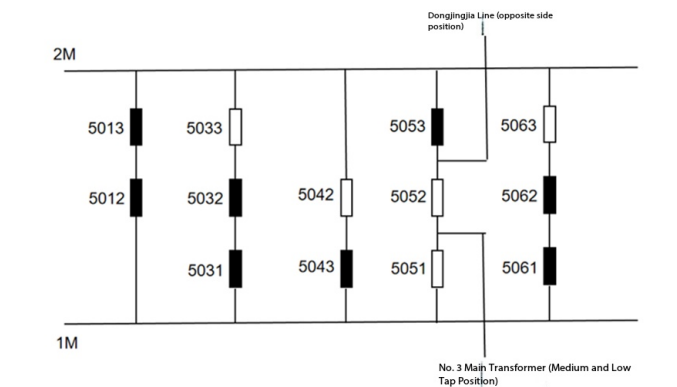
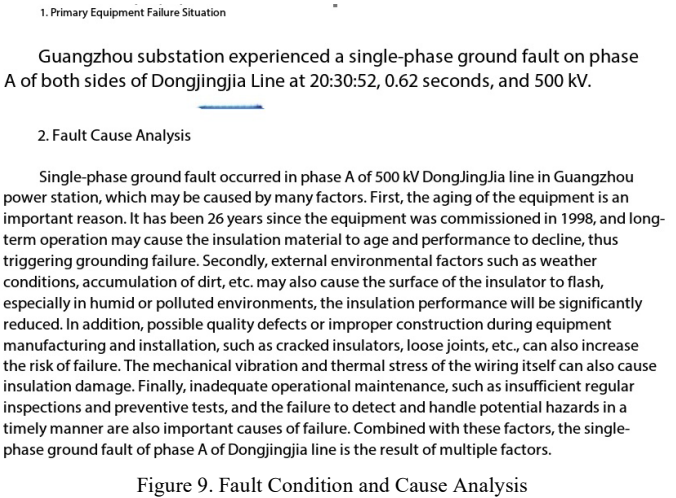


Figure 8. Pre-event Operating Mode

The GLM-4-Plus model, as the core engine, successfully generates event summaries containing key elements and fault cause analysis, event nature, exposed issues, and preventive suggestions, in accordance with industry standards. Through A/B testing and BLEU scores, the quality of model output has continuously improved.The fault conditions and cause analysis results generated by the model are shown in Figure 9.



The RAG knowledge-enhanced framework, in collaboration with GLM-4-Plus, solves the problem of insufficient domain knowledge, constructing a power grid-specific knowledge retrieval mechanism. In secondary system analysis, RAG+GLM-4-Plus successfully interpreted complex time-sequence data, judged protection conditions based on retrieved knowledge, and generated professional action behavior analysis, greatly enhancing accuracy and credibility.The preventive measures and recommendations as shown in Figure 10.

- To prevent similar incidents from recurring, the following measures are recommended:
1. Strengthen regular inspection and evaluation of old equipment to identify potential risks in advance.
 2. Improve equipment insulation performance, especially in humid environments, and strengthen protection for insulating materials.
 3. Improve environmental monitoring and analysis systems to promptly understand how changes in the environment affect equipment and take appropriate preventive measures. ✓
 4. Strengthen training for equipment management and maintenance personnel to enhance awareness and response capabilities regarding equipment aging and environmental factors.

Figure 10. Preventive Measures and Recommendations

The experiment results indicate that the system successfully generated complete fault event analysis reports, with professional content and accurate insertion of images and tables. Compared to manual report generation, the time for report preparation has been reduced to minutes, showing a significant increase in efficiency. This experiment fully validates the feasibility of the large model-based automated report generation system.

V. CONCLUSIONS

This study addresses the low efficiency and strong dependence on manual labor in traditional power fault report preparation by proposing an automated reporting method based on multi-source data fusion and a RAG-LLM collaborative generation framework. OCR and CNN techniques are utilized to recognize and structure multimodal data such as oscillograph waveforms and equipment logs, while the RAG framework enhances the accuracy and logical consistency of the LLM in professional power-system scenarios. Experimental results show that the proposed approach significantly reduces human involvement and outperforms conventional manual reporting in terms of accuracy, completeness, and consistency,

demonstrating its practical value for intelligent operation and maintenance of power grids.

Despite these advantages, the system still has room for improvement in handling non-standardized data, adapting to complex operational scenarios, and identifying faults in newly deployed equipment. Future work will focus on strengthening the system's real-time performance and generalization capability to better support the intelligent operational needs of modern power systems.

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